

Empirical Research in Finance

3. Interpreting regression output

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Outline

Estimation and Significance

- Statistical significance

- Economic significance

Goodness of fit

Non-linear specifications

- Transforming variables

- Polynomials

Estimation

Wooldridge dataset CEOSAL2: cross-sectional information on CEO salaries. (Notice units of variables!)

- ▶ salary: 1990 compensation, **USD thousands**
- ▶ sales: 1990 firm sales, **USD millions**
- ▶ ceoten: **years** as CEO with company

We run

$$\text{salary}_i = \beta_0 + \beta_1 \text{sales}_i + \beta_2 \text{ceoten}_i + u_i$$

```
> require(wooldridge)
> attach(ceosal2)
> m <- lm(salary ~ sales + ceoten)
> summary(ceosal2)
```

salary	age	college	grad
Min. : 100.0	Min. :33.00	Min. :0.0000	Min. :0.0000
1st Qu.: 471.0	1st Qu.:52.00	1st Qu.:1.0000	1st Qu.:0.0000
Median : 707.0	Median :57.00	Median :1.0000	Median :1.0000
Mean : 865.9	Mean :56.43	Mean :0.9718	Mean :0.5311
3rd Qu.:1119.0	3rd Qu.:62.00	3rd Qu.:1.0000	3rd Qu.:1.0000

Estimation in R

```
> summary(m)
```

```
Call:
```

```
lm(formula = salary ~ sales + ceoten)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-985.8	-306.4	-122.1	249.9	4405.6

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.217e+02	6.610e+01	9.405	< 2e-16 ***
sales	3.780e-02	6.668e-03	5.669	5.86e-08 ***
ceoten	1.393e+01	5.677e+00	2.453	0.0152 *

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 537.4 on 174 degrees of freedom
```

```
Multiple R-squared:  0.1732,          Adjusted R-squared:  0.1637
```

```
F-statistic: 18.22 on 2 and 174 DF,  p-value: 6.542e-08
```

Estimation in STATA

ssc install bcuse

bcuse ceosal2

reg salary ceoten sales

sum salary ceoten sales

```
. reg salary ceoten sales
```

Source	SS	df	MS	Number of obs =	177
Model	10522179.1	2	5261089.56	F(2, 174) =	18.22
Residual	50243785.6	174	288757.389	Prob > F =	0.0000
				R-squared =	0.1732
				Adj R-squared =	0.1637
Total	60765964.7	176	345261.163	Root MSE =	537.36

salary	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
ceoten	13.92561	5.677433	2.45	0.015	2.720113	25.13111
sales	.0378012	.0066679	5.67	0.000	.0246409	.0509615
_cons	621.6709	66.09745	9.41	0.000	491.215	752.1269

```
. sum salary ceoten sales
```

Variable	Obs	Mean	Std. Dev.	Min	Max
salary	177	865.8644	587.5893	100	5299
ceoten	177	7.954802	7.150826	0	37
sales	177	3529.463	6088.654	29	51300

Significance - Statistical significance

- ▶ A hypothesis test of the hypothesis $H_0 : \beta_j = b_j$:

$$t_j = \frac{\hat{\beta}_j - b_j}{\text{s.e.}(\hat{\beta}_j)}$$

- ▶ It's a measure of how strong our “signal” (the coefficient) is, compared to the “noise” (its uncertainty).
 - ▶ Big $t \rightarrow$ strong signal, small noise $\rightarrow$ we trust the estimate.
 - ▶ Small $t \rightarrow$ weak signal, large noise $\rightarrow$ could just be luck.
- ▶ Most software report the t_j for $b_j = 0$.
(Is the coefficient “statistically significant”?)
This t-statistic is approximately $\sim N(0, 1)$.

How Sure Are We? Understanding Significance Levels

Think of it like this: Before you “bet” your reputation on a finding, ask — “how sure am I this isn’t random noise?”

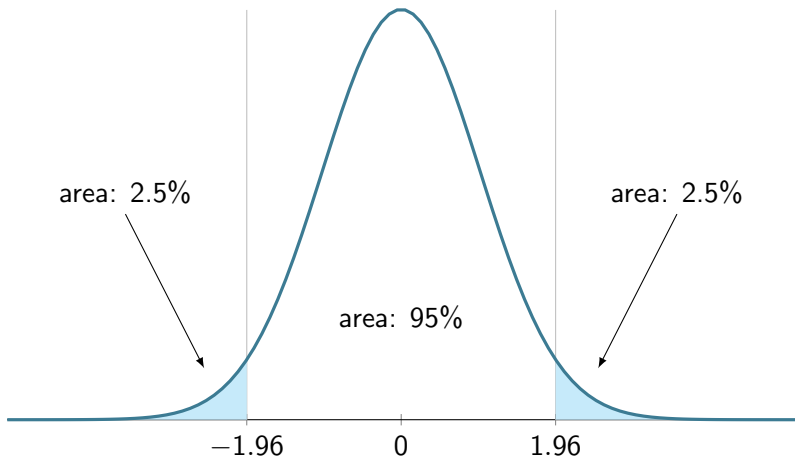
- ▶ 0.10: “I’m 90% sure.” — you’d “maybe” take a small bet
- ▶ 0.05: “I’m 95% sure.” — you’d easily bet (the academic standard)
- ▶ 0.01: “I’m 99% sure.” — you’d very easily bet

At 5% significance, the critical values for a two-sided test statistic for a normal distribution are $-1.96, +1.96$.

⇒ rule of thumb:

$$|t_j| > 1.96 \Rightarrow$$

“coefficient statistically different from zero at 5%.”



P-values

“P-value:” what is the **smallest** significance level at which H_0 would be rejected?

```
. reg salary ceoten sales
```

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p-values are less than 5%, since all H_0 are rejected at the 5% level.
one p-value is greater than 1%, then H_0 is not rejected at the 1% level.

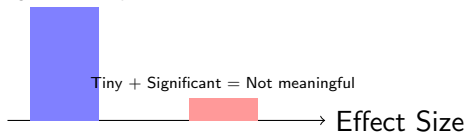
Does It Actually Matter? Economic Significance

Statistical significance asks: “Is this effect *real* or just random noise?”

Economic significance asks: “Even if it’s real... is it *big enough to matter*?”

- ▶ You find: High-ESG firms earn 0.05% more per year, and it’s statistically significant ($p < 0.01$).
- ▶ Sounds good... but 0.05%? That’s \$50 on a \$100,000 portfolio.
- ▶ It’s real — but who cares?

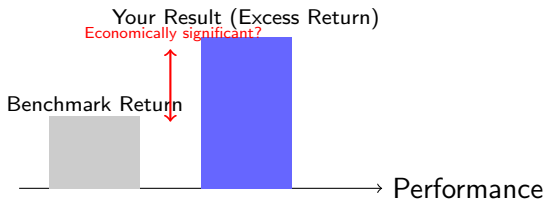
Big + Significant = Important



A result can be *real* (statistically significant) but *irrelevant* (economically tiny).

Significance - Economic significance

- ▶ Unlike with statistical significance, there are no cut-off points.
- ▶ The reader needs to be convinced that the results are meaningful.
 - ▶ compared to what? We need benchmarks.
 - ▶ Make use of descriptive statistics and compare with results from other studies.



Significance - Economic significance II

Going back to our CEO application...

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u$$

- ▶ (Intercept): $E[y|x_1 = 0, x_2 = 0] = \beta_0$
 - ▶ A CEO of a firm with 0 sales (out of sample!) with job tenure of 0 years has an average salary of \$621,670.
- ▶ sales: $E[\Delta y|\Delta x_1] = \beta_1 \Delta x_1$.
 - ▶ A CEO with 1 (million \$) more in sales ($\Delta x_1 = 1$) earns on average \$37.80 (0.037801 thousand) more (holding tenure fixed). (Is this big?)
 - ▶ A 1-standard-deviation increase in sales ($\Delta x_1 = 6,088.654$), increases salary on average by \$230,158 (holding tenure fixed)
 $\implies$ 26.6% of average salary.
- ▶ ceoten: $E[\Delta y|\Delta x_2] = \beta_2 \Delta x_2$.
 - ▶ A CEO with one additional year on the job ($\Delta x_2 = 1$) earns on average 13,925 USD more (**holding sales fixed**). (Is this big? Repeat as above.)

Goodness of fit

- ▶ The R^2 measures how much variation of y is explained by the regressors.
 - ▶ Cross-sectional regressions often have low R^2 s
- ▶ The “adjusted R^2 ” is similar, but decreases when you add regressors that do not help much in explaining y .
- ▶ The F -test tests for joint significance of all regressors:
Is $\beta_1 = \beta_2 = \dots = 0$?

Non-linear specifications

“Linear regression” only means “linear in the coefficients.”

In an OLS specification,

- ▶ we cannot have $\log(\beta_1)$, β_3^2 , ...
- ▶ we can have x^2 , $\log(y)$, ...

It is harder to interpret coefficient estimates in non-linear OLS specifications.

Transforming variables

Model	Dependent variable	Independent variable	Interpretation of β
Level-level	y	x	$\Delta y = \beta \Delta x$
Log-level	$\ln(y)$	x	$\% \Delta y \approx 100 \beta \Delta x$
Level-log	y	$\ln(x)$	$\Delta y \approx \frac{\beta}{100} \% \Delta x$
Log-log	$\ln(y)$	$\ln(x)$	$\% \Delta y = \beta \% \Delta x$

- ▶ Log of variable = relative changes
- ▶ Common in finance: log returns, log market cap

Transforming variables in R

```
> lsalary <- log(salary)
> lsales <- log(sales)
>
>
>
> m <- lm(lsalary ~ sales + ceoten)
> m <- lm(salary ~ lsales + ceoten)
> m <- lm(lsalary ~ lsales + ceoten)
```

Transforming variables in Stata

```
gen lsalary = log(salary)  
gen lsales = log(sales)
```

```
reg lsalary sales ceoten
```

```
reg salary lsales ceoten
```

```
reg lsalary lsales ceoten
```

VARIABLES	(1) lsalary	(2) salary	(3) lsalary
sales	0.00004*** (6.1285)		
ceoten	0.01213** (2.0909)	13.10171** (2.3673)	0.01143** (2.1237)
lsales		179.61449*** (6.4995)	0.22643*** (8.4237)
Constant	6.33896*** (93.8469)	-537.15405** (-2.5582)	4.85458*** (23.7694)
Observations	177	177	177
R-squared	0.188	0.212	0.299

t-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

- ▶ A CEO with one additional year on the job earns on average 1.213% more.
- ▶ A CEO of a company with 1% higher sales earns on average 1,796.144 USD more.
- ▶ A CEO of a company with 1% higher sales earns on average 0.22643% more.

VARIABLES	(1) lsalary	(2) salary	(3) lsalary
sales	0.00004*** (6.1285)		
ceoten	0.01213** (2.0909)	13.10171** (2.3673)	0.01143** (2.1237)
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When to take logs

- ▶ A better characterization of how salary changes with sales and ceo tenure is perhaps expressed as a percentage.
- ▶ **Rule of thumb:** take logs if a variable varies over a very wide range (e.g., firm size, sales, income) - it compresses the scale, making the relationship more linear and easier to interpret.
- ▶ If a variable contains zeroes, one could use $\ln(1+y)$ instead of $\ln(y)$.
- ▶ Careful: The intercept in these models has little meaning.

Polynomials

- ▶ Capture increasing or decreasing marginal effects.
- ▶ We can include several transformations of the same variable, which allows for polynomials. E.g.:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{1i}^2 + u$$

- ▶ Interpreting coefficients becomes more difficult:

Since $\frac{\partial y}{\partial x_1} = \beta_1 + 2\beta_2 x_1$, we have $\Delta y \approx (\beta_1 + 2\beta_2 x_1) \Delta x_1$.

- ▶ Typically, plug in the **average** value of x_1 in the sample, or some other interesting value (median, lower or upper quartile values).

A Variable Can Be Important Even If It Is Not Significant

Common mistake: “If a variable is not statistically significant, we should remove it.”

Why this is not always correct:

- ▶ **Theory matters:** A variable may be important according to finance theory.
- ▶ **Controls matter:** A variable can help us isolate the effect of the main explanatory variable.
- ▶ **Insignificant does not mean irrelevant:** It may only mean the sample is small or noisy.

A high p-value means:

“We do not have strong evidence in this sample.”

It does **not** automatically mean:

“This variable has no effect and should be deleted.”

Example: Why Keeping a Control Variable Can Matter

Suppose we want to estimate the effect of **leverage** on **firm returns**.

We also include **market volatility** in the model.

- ▶ In our sample, volatility is **not statistically significant**.
- ▶ But volatility may still affect **returns**.
- ▶ Volatility may also be related to **leverage**.

What happens if we remove volatility?

Then part of the effect of volatility may be wrongly captured by the leverage coefficient.

Result: The estimated effect of leverage can become **biased**.

Important: This Shows Correlation, Not Causality

Our regression shows that **leverage and returns are associated**, after controlling for volatility.

What does it not show?

It does **not** automatically prove that leverage **causes** returns.

Why not? For example:

- ▶ There may be **other omitted variables** affecting both leverage and returns.
- ▶ The direction of influence may be unclear (**reverse causality**).

Reading:

- ▶ Wooldridge, chapters 1-6.